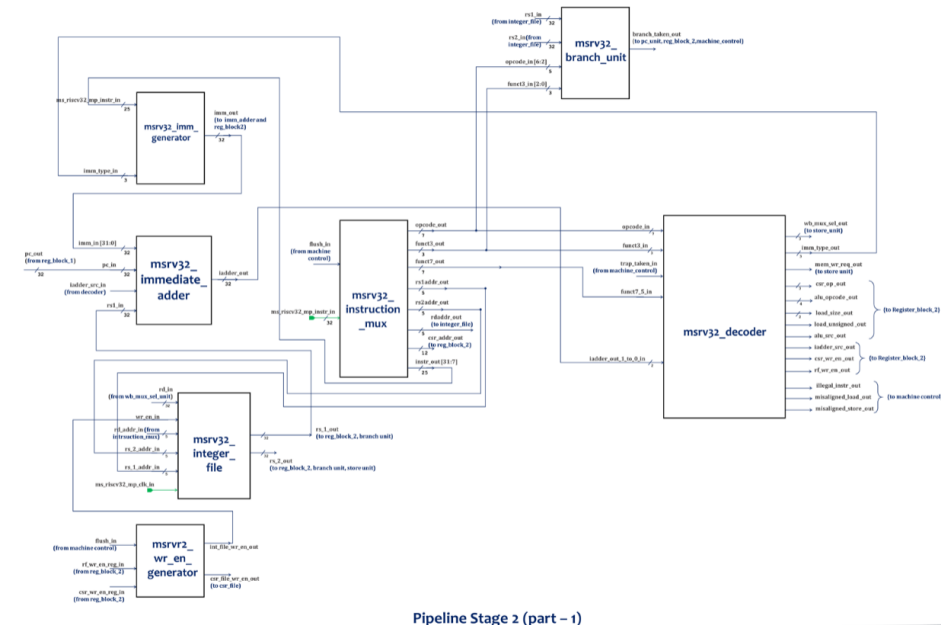
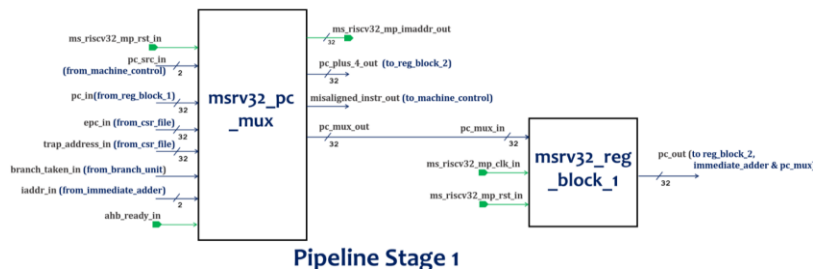
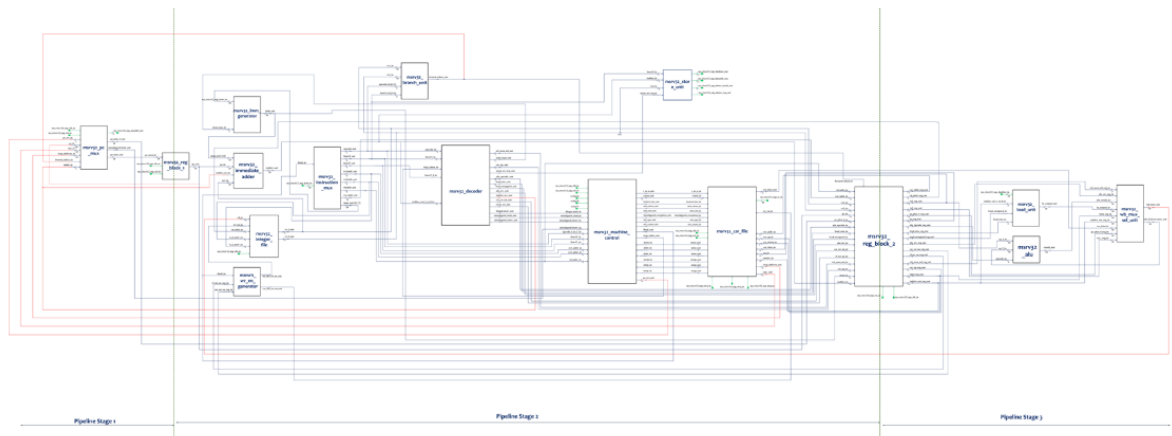


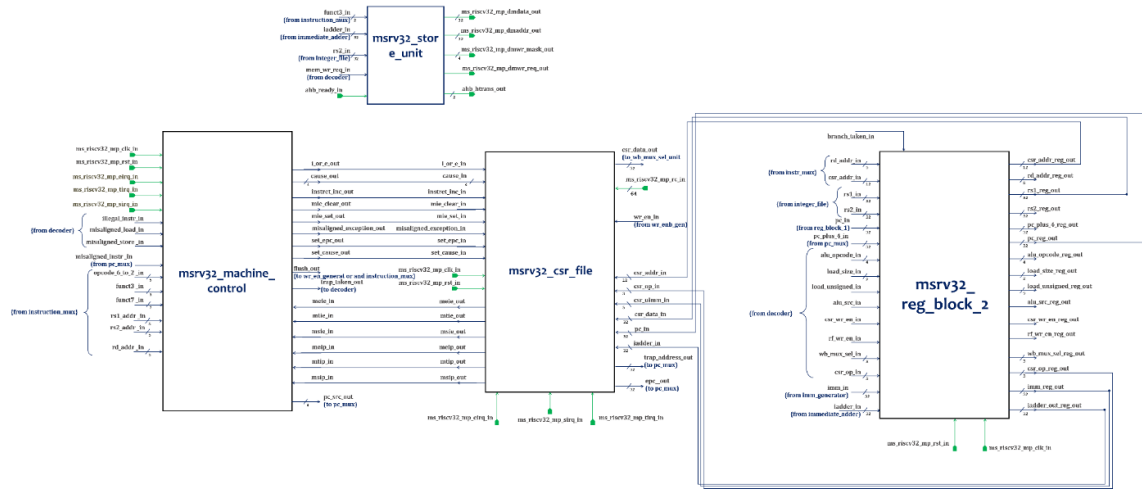
Maven Silicon VIT_DI09 RISC-V

Internship Report

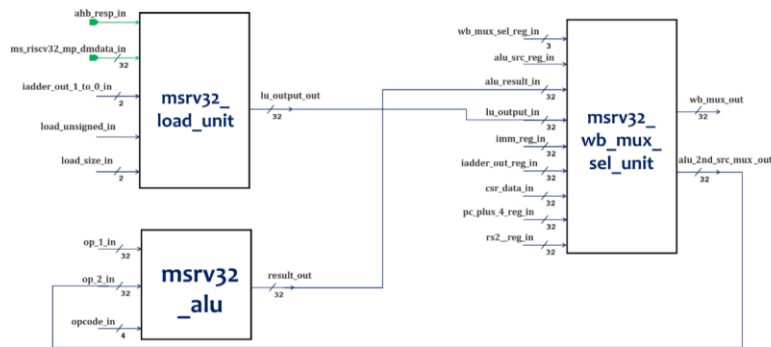
Name: Sreevignesh Balamurugan

Block Diagrams:





Pipeline Stage 2 (part - 2)



Pipeline Stage 3

Simulation instruction:

```
[SreevigneshB@mavenserver-RH4 sim]$ make run_r_type
```

Result:

```
#
# -----
# 8888888b.      d8888      .d8888b.      .d8888b.
# 888   Y88b     d888888 d88P  Y88b d88P  Y88b
# 888   888     d88P888 Y88b.   Y88b.
# 888 d88P d88P 888 *Y888b.   *Y888b.
# 8888888P* d88P 888 *Y88b.   *Y88b.
# 888      d88P 888 *888      *888
# 888      d8888888888 Y88b d88P Y88b d88P
# 888      d88P 888 *Y8888P* *Y8888P*
# -----
# ----- NOP or NO_INSTRUCTION -----
# UVM_INFO verilog_src/uvm-1.1d/src/base/uvm_objection.svh(1267) @ 1000091500: reporter [TEST_DONE] 'run' phase is ready to
# oceed to the 'extract' phase
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :60008
# UVM_WARNING : 1
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [IN_AGENT TOP] 1
# [Questa UVM] 2
# [RNTST] 1
# [TEST_DONE] 1
# [UVMTOP] 1
# [r_type_sequence] 10000
# [uvm_sequence_base] 1
# [uvm_test_top.env.instr_agt_for_duv.drv] 1
# [uvm_test_top.env.instr_agt_for_duv.seqr.r_type_seq1] 50001
# ** Note: $finish : /home/cad/eda/MENTORGRAPHICS/Questasim/questasim/linux_x86_64/./verilog_src/uvm-1.1d/src/base/uvm_r
# t.svh(430)
# Time: 1000091500 ps Iteration: 54 Instance: /msrv32_tb_top
# Saving coverage database on exit...
```

Files:

```
Errors: 0, Warnings: 0
[SreevigneshB@mavenserver-RH4 sim]$ cd ../rtl
[SreevigneshB@mavenserver-RH4 rtl]$ ls
csr_data_mux_unit.v      mie_reg.v               msrv32_decoder.v        msrv32_machine_control.v  msrv32_top_wrapper.sv
data_wr_mux_unit.v      mip_reg.v               msrv32_imm_generator.v  msrv32_pc.v               msrv32_wb_mux_sel_unit.v
machine_counter.v       misa_and_pre_data.v     msrv32_immediate_adder.v msrv32_reg_block2.v       msrv32_wr_en_generator.v
machine_counter_setup.v msrv32_alu.v            msrv32_instruction_mux.v msrv32_reg_block1.v       mstatus_reg.v
mcause_reg.v            msrv32_branchunit.v    msrv32_integer_file.v   msrv32_store_unit.v       mtval_reg.v
mepc_and_mscratch_reg.v msrv32_csr_file.v       msrv32_load_unit.v      msrv32_top.v              mtvec_reg.v
```